**Chapter 8, Lecture 10**

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## Bootstrap

Consider $X_1, \ldots, X_n$ iid $f(x|\theta_0)$. Suppose we have an estimator

$$\widehat{\theta} = \widehat{\theta}(X_1, \ldots, x_n).$$

If we knew the true parameter value $\theta_0$, we could take many samples, and approximate the sampling distribution of $\widehat{\theta}$, just like we did in our simulations. The *parametric* bootstrap does something similar. We draw instead an iid *bootstrap* sample $X_1^*, \ldots, X_n^*$ from $f(x|\widehat{\theta})$, and compute the bootstrap estimate

$$\widehat{\theta}^* = \widehat{\theta}(X_1^*, \ldots, X_n^*)$$

Same formula, now based on the bootstrap sample. We imitate in the bootstrap world the real world, and it can be proved in some generality that $\widehat{\theta} - \theta$ has approximately the same (conditional) distribution as $\widehat{\theta}^* - \widehat{\theta}$ (given the data $X_1, \ldots, X_n$). The latter we know in principle, although, again, it's easy to approximate this distribution by taking many bootstrap samples, and approximating the distribution of $\widehat{\theta}^* - \widehat{\theta}$. This implies that, taking $\delta_*$ to be the $1 - \alpha/2$-upper point of $\widehat{\theta}^* - \widehat{\theta}$, and $\delta^*$ to be the $\alpha/2$-upper point of $\widehat{\theta}^* - \widehat{\theta}$,

$$\begin{aligned} 1 - \alpha &\approx \mathbb{P}^*\{\delta_* \leq \widehat{\theta}^* - \widehat{\theta} \leq \delta^*\} \\ &\approx \mathbb{P}\{\delta_* \leq \widehat{\theta} - \theta \leq \delta^*\} \end{aligned}$$

Can you explain the two approximations? A simple algebra yields the $(1 - \alpha)$ – bootstrap confidence interval

$$[\widehat{\theta} - \delta^*, \widehat{\theta} - \delta_*].$$

The bootstrap estimates the sampling distribution of the estimator.

EXAMPLE. Suppose $X_1, \ldots, X_n$ are iid $N(\mu, \sigma^2)$. Take $\widehat{\mu} = \overline{X}_n$ and $S^2$ as estimates for $\mu$ and $\sigma^2$. We are interested in a bootstrap confidence interval for $\mu$. Following the recipe above, draw an iid sample $X_1^*, \ldots, X_n^*$ from $N(\overline{X}, S^2)$. In this case, we actually know the conditional distribution of $\overline{X}^*$ - it is $N(\overline{X}, S^2/n)$. Hence, $\overline{X}^* - \overline{X} \sim N(0, S^2/n)$, and

$$\delta_* = -\frac{S}{\sqrt{n}} z(\alpha/2), \qquad \text{and} \qquad \delta^* = \frac{S}{\sqrt{n}} z(\alpha/2),$$

resulting in the bootstrap CI

$$\overline{X} \pm \frac{S}{\sqrt{n}} z(\alpha/2)$$

This is asymptotically correct. Although not mentioned in the textbook, there is a better approach, namely to use Studentized statistics. Just as before, it is reasonable to asume, and this can be proved in many cases, that

$$\frac{\widehat{\theta} - \theta}{s(\widehat{\theta})}$$

has approximately the same distribution, for $n$ large, as

$$\frac{\widehat{\theta}^* - \widehat{\theta}}{s(\widehat{\theta}^*)}$$

Here $s(\widehat{\theta})$ is an estimate of the standard error of $\widehat{\theta}$.[1] If $\delta^*$ is the $\alpha/2$ upper point of the distribution of $(\widehat{\theta}^* - \widehat{\theta})/s(\widehat{\theta}^*)$, we obtain

$$\begin{aligned} 1 - \alpha &\approx \mathbb{P}^* \left\{ \delta_* \leq \frac{\theta^* - \widehat{\theta}}{S(\widehat{\theta}^*)} \leq \delta^* \right\} \\ &\approx \mathbb{P}\{ \delta_* \leq \frac{\widehat{\theta} - \theta}{S(\widehat{\theta})} \leq \delta^* \} \end{aligned}$$

and we obtain the interval

$$[\widehat{\theta} - \delta^* S(\widehat{\theta}), \widehat{\theta} - \delta_* S(\widehat{\theta})]$$

[1]For instance, if we have a sample from an Exponential($\lambda$) distribution, $\widehat{\lambda} = 1/\overline{X}_n$, our bootstrap sample is drawn from Exponential($\widehat{\lambda}$), and $s(\widehat{\lambda}) = \widehat{\lambda}/\sqrt{n}$ or you can take $s(\widehat{\lambda}) = (n/(n-1))\widehat{\lambda}/\sqrt{n-2}$.

EXAMPLE We continue with the example above. Using the same bootstrap sample $X_1^*, \ldots, X_n^*$ iid $N(\overline{X}_n, S^2)$, we see that the Studentized statistic

$$\frac{\overline{X}_n^* - \overline{X}_n}{S^*/\sqrt{n}}$$

with

$$\overline{X}_n^* = \frac{1}{n}\sum_{i=1}^n X_i^* \quad \text{and} \quad S^* = \sqrt{\frac{1}{n-1}\sum_{i=1}^n (X_i^* - \overline{X}_n^*)^2}$$

has exactly the same distribution as

$$\frac{\overline{X}_n - \mu}{S/\sqrt{n}}$$

as both have a Student $t_{n-1}$ distribution. This means that the bootstrap upper points

$$\delta_* = t_{n-1}(1-\alpha/2) = -t_{n-1}(\alpha/2), \qquad \delta^* = t_{n-1}(\alpha/2)$$

and we arrive at the bootstrap interval

$$[\widehat{\theta} - \delta^* S(\widehat{\theta}), \widehat{\theta} - \delta_* S(\widehat{\theta})] = [\overline{X}_n \pm \frac{S}{\sqrt{n}} t_{n-1}(\alpha/2)]$$

This is the same as the exact confidence interval for $\mu$ derived before. We see in this example that it pays off to use Studentized statistics.

Typically when people refer to the bootstrap, they refer to the *nonparametric* bootstrap. This is similar in spirit, but we draw with replacement from the original sample. Generally the bootstrap samples are more accurate in that the claimed coverage is more accurate, at the cost of slightly larger intervals. There is no free lunch.

We will run some simulations to convey this concept.

EXAMPLE: WHICH CONFIDENCE BANDS ARE BETTER? Suppose $X_1, \ldots, X_n$ iid Exponential with parameter $\lambda$. Then $\widehat{\lambda} = 1/\overline{X}$ and recall that the asymptotic variance is $\lambda^2/n$. We want to compare three confidence intervals:

- Asymptotic confidence interval based on the asymptotic normaility of the MLE: simply compute
$$\left[\widehat{\lambda} \pm z(\alpha/2)\frac{\widehat{\lambda}}{\sqrt{n}}\right]$$
- Bootstrap version 1: Draw $X_1^*, \ldots, X_n^*$ iid from an Exponential with parameter $\widehat{\lambda}$. Compute $\lambda^* = 1/\overline{X}^* = n/\sum_{i=1}^n X_i^*$. Repeat this many (10,000) times to compute $\delta_*, \delta^*$ such that
$$\mathbb{P}^*\left\{\delta_* \leq \widehat{\lambda}^* - \widehat{\lambda} \leq \delta^*\right\} = 1 - \alpha.$$
The $1-\alpha$ bootstrap confidence interval is
$$\left[\widehat{\lambda} - \delta^*, \widehat{\lambda} - \delta_*\right]$$
- Bootstrap version 2: Draw $X_1^*, \ldots, X_n^*$ iid from an Exponential with parameter $\widehat{\lambda}$. Compute $\lambda^* = 1/\overline{X}^* = n/\sum_{i=1}^n X_i^*$. Repeat this many (10,000) times to compute $\varepsilon_*, \varepsilon^*$ such that
$$\mathbb{P}^*\left\{\varepsilon_* \leq \frac{\widehat{\lambda}^* - \widehat{\lambda}}{\widehat{\lambda}^*/\sqrt{n}} \leq \varepsilon^*\right\} = 1 - \alpha.$$
The $1-\alpha$ (Studentized) bootstrap confidence interval is
$$[\widehat{\lambda} - \varepsilon^*\frac{\widehat{\lambda}}{\sqrt{n}}, \widehat{\lambda} - \varepsilon_*\frac{\widehat{\lambda}}{\sqrt{n}}]$$

How do these procedures rank in terms of coverage? And in length? In order to answer this, run various simulations. Consider $\lambda = 2$, $\alpha = 0.05$ and $n = 50$. To reduce variability due to the simulations, each time you have generated the bootstrap sample, compute $\delta_*, \delta^*, \varepsilon_*, \varepsilon^*$, all on the same bootstrap sample. (If you've time, also try $n = 25$ and $n = 100$.)